

ERA5 DATASET ONBOARDING NOTES

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Target Dataset:	ERA5 Reanalysis (ECMWF)	Document Type:	Technical Onboarding & Architecture Spec
Target Variable:	Total Precipitation (tp)	Target Grain:	State × Day (Aggregated from 0.25° Grid)

1. Strategic Overview: What is ERA5 & Why Choose It?

What is ERA5 and Who Produces It?

ERA5 is the fifth generation atmospheric reanalysis of the global climate, produced by the **European Centre for Medium-Range Weather Forecasts (ECMWF)** through the Copernicus Climate Change Service (C3S). A "reanalysis" is not raw satellite telemetry or sparse weather station data. It is a modern hindcast that blends historical observations (from satellites, radar, weather balloons, and ground stations) with a rigid physical weather forecasting model using advanced 4D-Var data assimilation techniques. This fills data gaps globally, providing a comprehensive, spatially continuous estimate of atmospheric parameters across decades.

Why Did We Choose ERA5 over Alternatives?

For insurance-grade catastrophe modeling, raw rain gauge networks (e.g., local meteorological stations) suffer from missing records, geographic clustering near urban centers, and shifting baselines. ERA5 resolves these issues by offering:

- **Perfect Spatial and Temporal Continuity:** Zero missing grid cells or unrecorded days across our complete multi-decade baseline window.
- **Rigid Standardization:** Consistent global grid sizing, removing the data-cleaning variability typically required when combining fragmented station networks.
- **Actuarial Integrity:** Maintained by a top-tier international scientific body (ECMWF), satisfying the data provenance standards required by corporate reinsurers and risk regulators.

2. Scientific Data Ingestion: The NetCDF File Format

What is NetCDF?

NetCDF (Network Common Data Form) is a self-documenting, array-oriented binary file format designed specifically for multi-dimensional geospatial data. Unlike plain flat files, a NetCDF file contains internal metadata describing its exact spatial projection, variable definitions, measurement units, and dimension parameters.

Why Doesn't ERA5 Come as a CSV?

Distributing global climate data via flat files like CSV is structurally impossible due to the exponential growth of multi-dimensional arrays. To represent a single variable covering a 40-year window at a high resolution, a CSV would require billions of rows. Every single line would have to explicitly repeat string values for latitude, longitude, and timestamps, creating a massive, highly inefficient storage bottleneck.

NetCDF avoids this layout via an array-native binary framework. Instead of repeating coordinate descriptions across lines, coordinates are defined exactly once along independent geometric paths (Dimensions). The actual physical observations are then stored within a highly compressed, multi-dimensional array (Data Cube).

The Geospatial 3D Data Cube Structure

Dimensions: Time × Latitude × Longitude

Every coordinate point maps directly to an array offset, enabling fast spatial sub-setting without scanning the full file.

3. Variable Strategy: Version 1 Scope & Rationale

ERA5 catalogs thousands of atmospheric, oceanic, and terrestrial variables. To minimize ingestion complexity while maximizing predictive capabilities, Version 1 limits the pipeline to a single meteorological driver.

Variable (Short Code)	Type	Version 1 Status	Actuarial Engineering Purpose & Scope Justification
Total Precipitation (tp)	Meteorological	✅ Ingested	The core physical driver. Sufficient to build rolling rainfall totals, historical anomalies, extreme thresholds, and GEV distributions.
Convective Precipitation (cp)	Meteorological	❌ Excluded	Isolates convective storms (thunderstorms). Extraneous for Version 1, as convective totals are already captured within tp.
Large-scale Precipitation (lsp)	Meteorological	❌ Excluded	Isolates synoptic frontal weather systems. Already accounted for within the accumulated tp variable.
Surface Runoff (sro)	Hydrological	⌚ Deferred (V2)	Measures water flowing over land. Excluded to avoid early structural complexity; proxied instead via rolling accumulation indices.
Volumetric Soil Water (swv11)	Terrestrial	⌚ Deferred (V2)	Measures topsoil moisture saturation. Highly useful, but deferred to keep initial processing lean. Proxied via consecutive wet days.

Critical Unit Warning (The Beginner's Trap):

ERA5 stores Total Precipitation (tp) as an accumulated depth in **meters (m)** per processing interval, rather than the standard meteorological measure of **millimeters (mm)**.

Underwriting Impact: Omitting the necessary conversion step ($\text{Value} \times 1000$) will scale calculated risk parameters down by a factor of 1,000, rendering any subsequent EVT calculations or flood risk thresholds entirely invalid. The engineering pipeline must explicitly include an element-wise conversion step.

4. NetCDF Technical Structure & Internal Metadata

When inspecting the downloaded NetCDF binaries using programmatic tools (e.g., Python's `xarray` or `netCDF4` library), the file reveals an internal structural layout organized across four primary vectors:

Dimensions:

```
time: 14610          # Daily entries across a continuous 40-year baseline
latitude: 121        # 0.25-degree increments across spatial bounds (e.g., 5°N to 35°N)
longitude: 161       # 0.25-degree increments across spatial bounds (e.g., 65°E to 105°E)
```

```

Coordinates:
  * time          (time)          datetime64[ns] 1980-01-01 ... 2019-12-31
  * latitude      (latitude)      float32 35.0 34.75 34.5 ... 5.0
  * longitude     (longitude)     float32 65.0 65.25 65.5 ... 105.0

Data Variables:
  tp              (time, latitude, longitude) float32 ... # The 3D physical data array

Attributes:
  title:          "ERA5 Reanalysis Daily Accumulations for Indian Subcontinent"
  institution:    "ECMWF / Copernicus Climate Change Service"
  history:        "Created via CDS API extraction script on 2026-07-02"

```

5. Expected Engineering Pipeline: Cube to Table Transformation

To transform this high-dimensional binary array into our target [One Row = One State × One Day] flat table, the data engineering pipeline must execute a structured, sequential workflow:

Spatial & Temporal Extraction Pipeline:

- Ingestion & Scale Normalization:** Programmatically mount the NetCDF binary files and apply an immediate element-wise multiplication by 1,000 to convert the raw data array from meters (m) to millimeters (mm).
- Administrative Shapefile Masking:** Ingest an official vectorized GIS shapefile containing Indian state administrative boundaries (e.g., local state divisions).
- Spatial Intersect & Zonal Aggregation:** Run a spatial intersection mapping every 0.25° ERA5 grid point to its containing administrative polygon. For each individual day, compute the spatial mean across all grid cells falling inside a state's boundaries:

State_Rainfall[t] = Mean(Grid_Rainfall[g, t] for all g inside State_Polygon)
- Temporal Feature Compounding:** Flatten the output into a database structure, sorting by state and calendar date. Compute the rolling historical metrics, continuous wet-day counts, long-term anomalies, and GEV-fitted risk distributions.
- Target Event Matching:** Join the structured records against the frozen EM-DAT flood history table to flag days where localized rainfall triggered recorded financial losses.

This pipeline successfully bridges the gap between raw physical atmospheric sciences and structured corporate risk modeling, turning unstructured data cubes into a specialized Hazard Master Table optimized for underwriting analytics.